

Experiment Report

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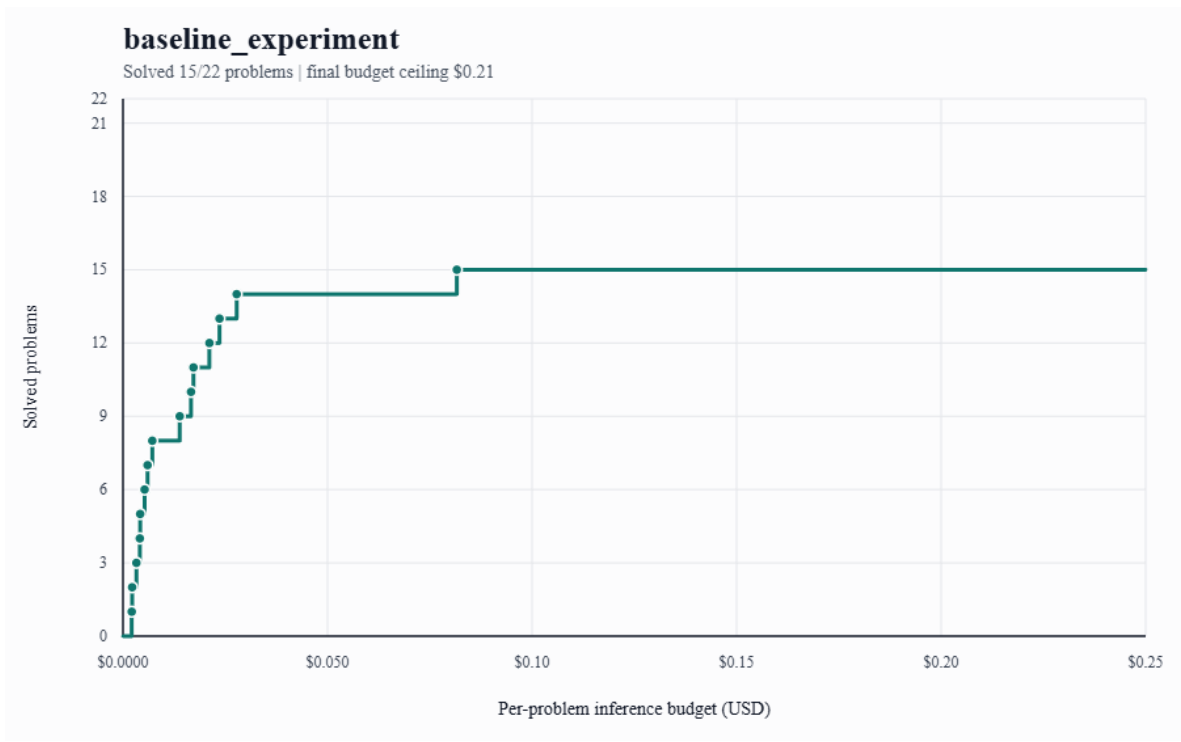
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New Strategy.....	Error! Bookmark not defined.

Baseline Experiments

First, test all baseline with newest model and check reasoning effort

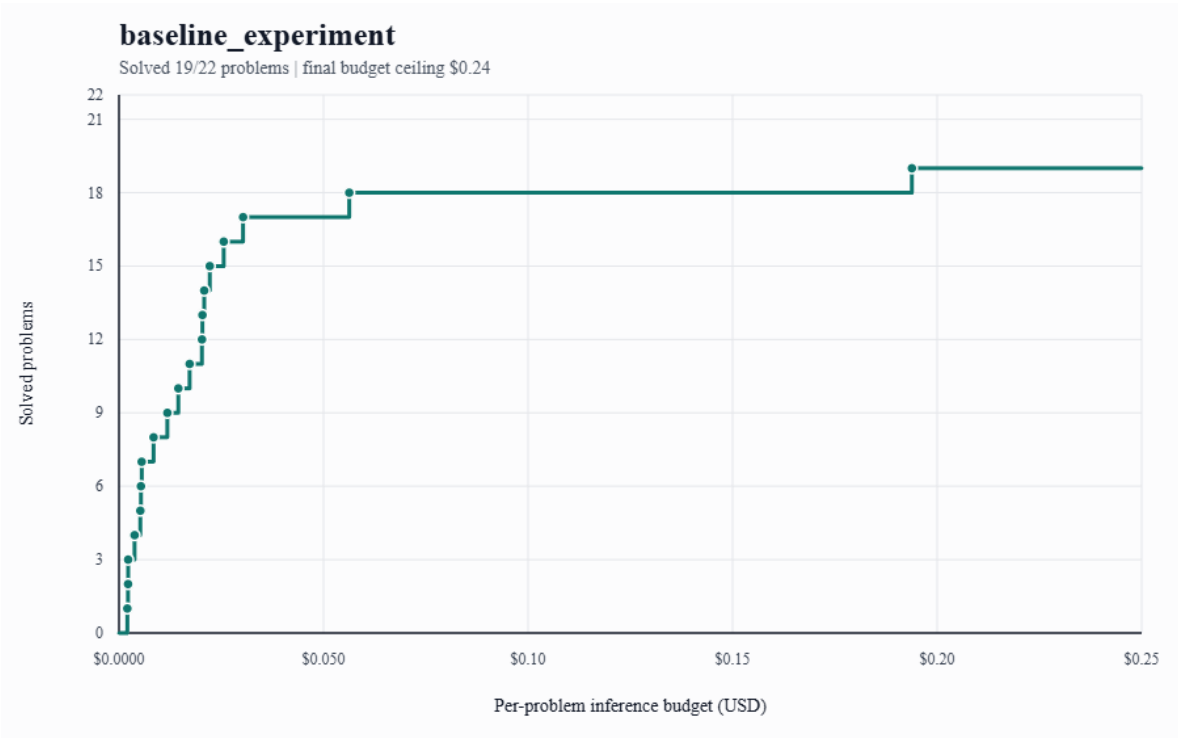
Baseline Experiment 1

Model	5.4	Correct	15
Max_budget	0.2	Total cost	1.6656
Max_loops	10		
Resoning effort	None		



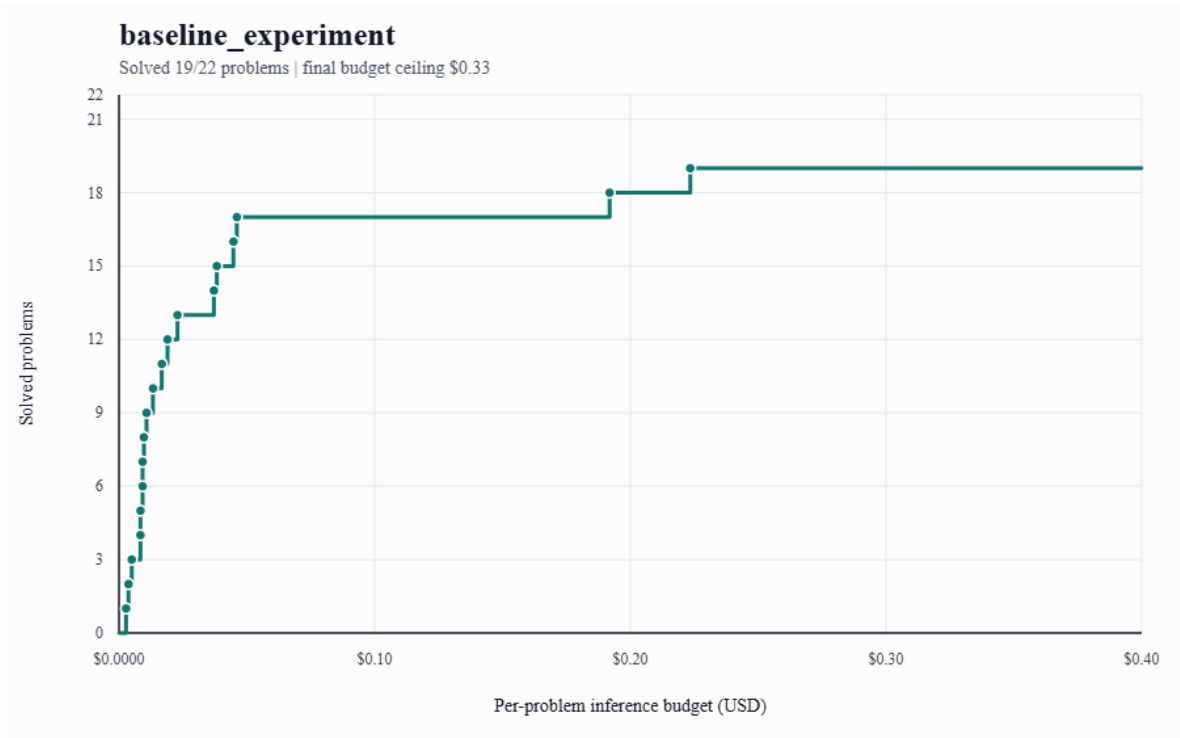
Baseline Experiment 2

Model	5.4	Correct	19
Max_budget	0.2	Total cost	1.13
Max_loops	10		
Resoning effort	low		



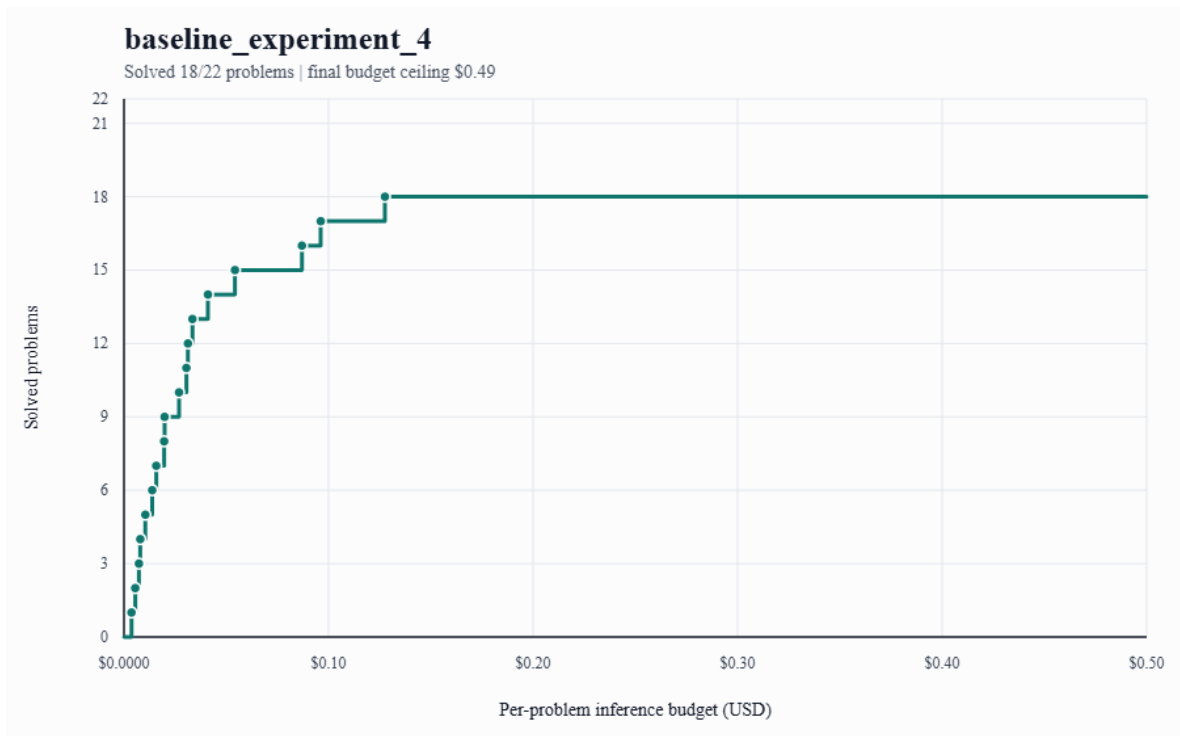
Baseline Experiment 3

Model	5.4	Correct	19
Max_budget	0.2	Total cost	1.4732
Max_loops	10		
Resoning effort	medium		



Baseline Experiment 4

Model	5.4	Correct	18
Max_budget	0.2	Total cost	1.83
Max_loops	10		
Resoning effort	high		



Baseline Experiment 5

Model	5.4	Correct	17
Max_budget	0.2	Total cost	3.164
Max_loops	10		
Resoning effort	xhigh		

Insight – baseline & reasoning

About reasoning effort. Taken that 5.4 is used:

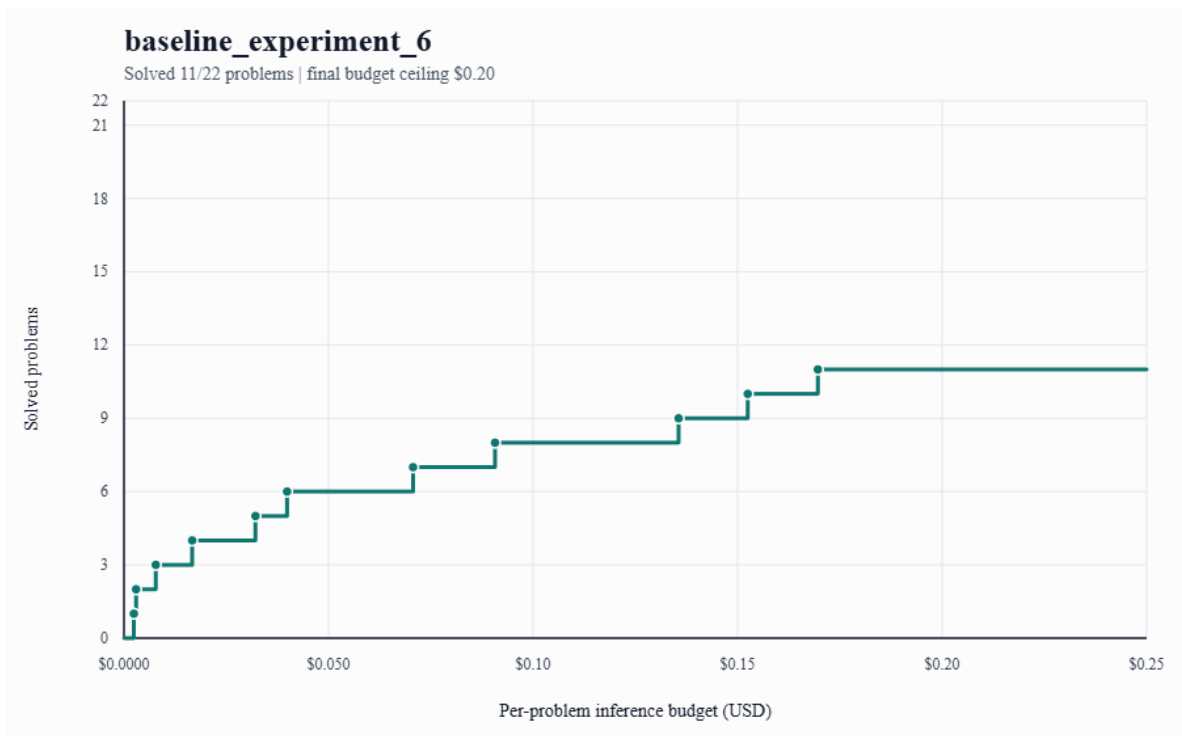
Low>medium> high > none > xhigh

Makes sense. Making the model force itself a little bit (default is set to none), for constructing the proofs should improve on the results. On the other hand as the proofs themselves are not that complicated, using “high or xhigh”.

Next -> study behavior in mini & nano

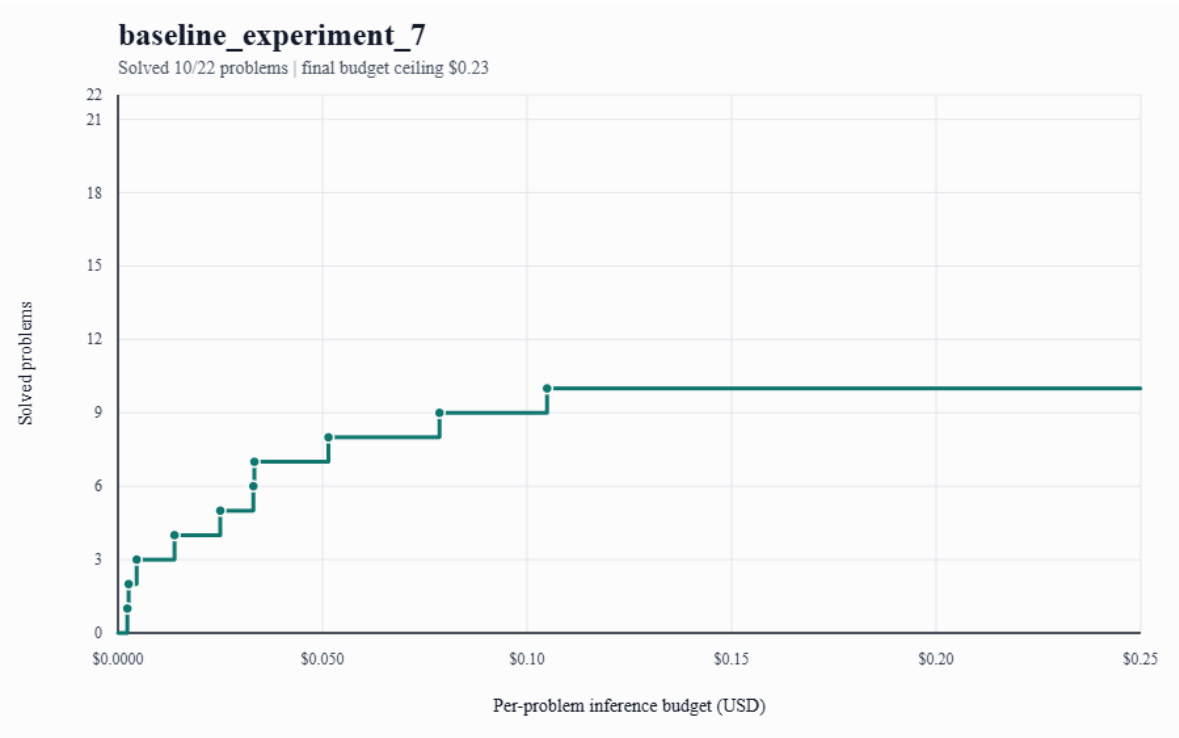
Baseline Experiment 6

Model	5.4-mini	Correct	11
Max_budget	0.2	Total cost	2.94
Max_loops	10		
Resoning effort	none		



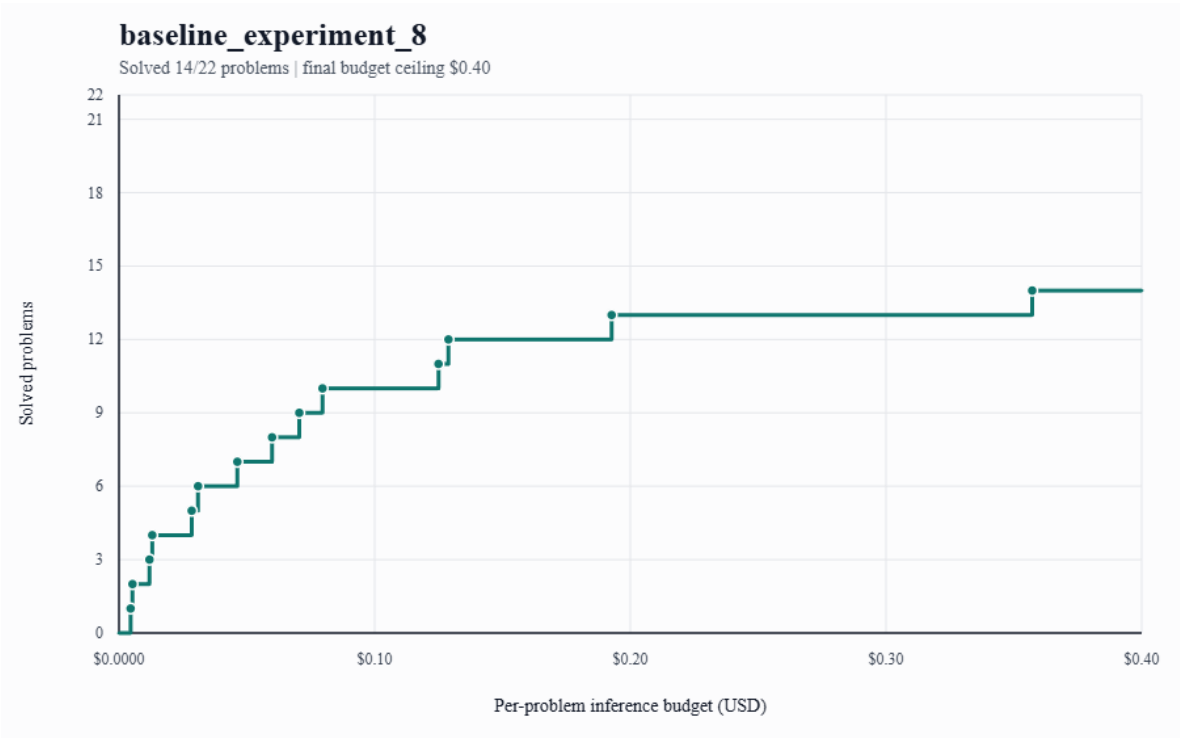
Baseline Experiment 7

Model	5.4-mini	Correct	10
Max_budget	0.2	Total cost	2.87
Max_loops	10		
Resoning effort	low		



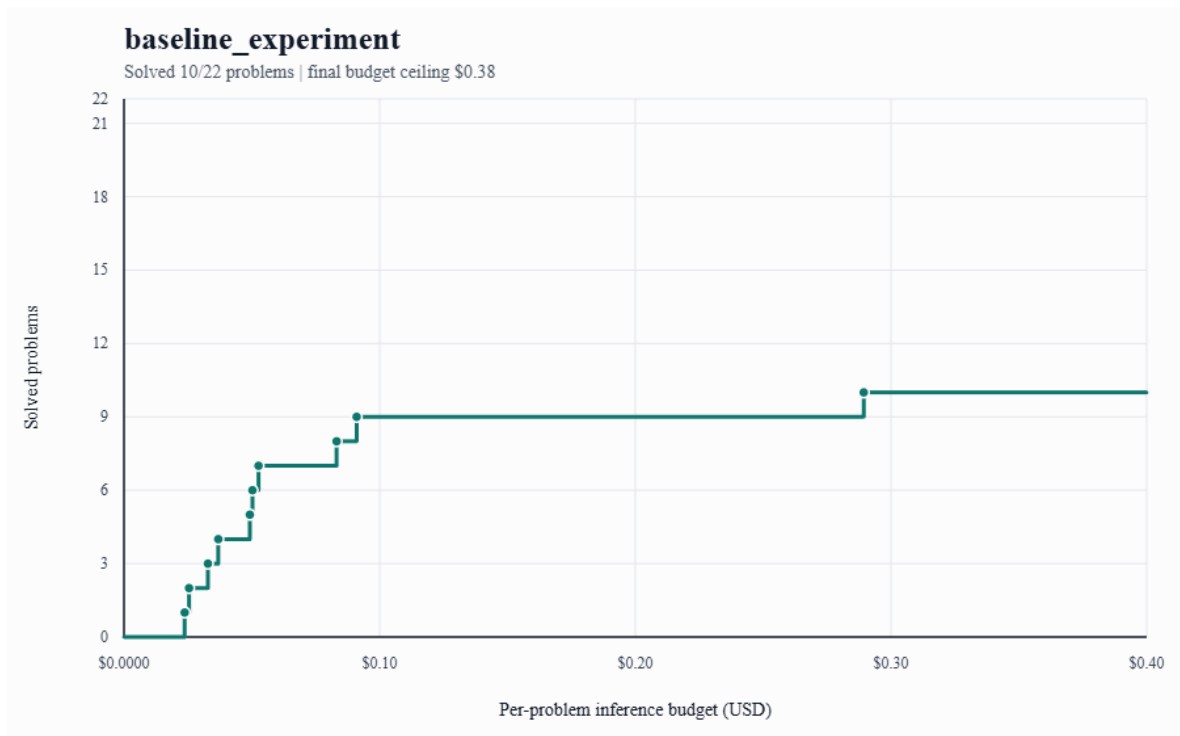
Baseline Experiment 8

Model	5.4-mini	Correct	14
Max_budget	0.2	Total cost	3.38
Max_loops	10		
Resoning effort	medium		



Baseline Experiment 9

Model	5.4-mini	Correct	10
Max_budget	0.2	Total cost	3,95
Max_loops	10		
Resoning effort	high		



Insight - Std vs mini

15 at 1,6\$ vs 11 at 2,9 \$

19 at 1,13\$ vs 10 at 1,8 \$

19 at 1,4\$ vs 14 at 3,3 \$

From these results we observe that for the baseline, it does not make sense to use mini over std 5.4. It does not offer a budget advantage or a performance advantage.

Insight – reasoning effort

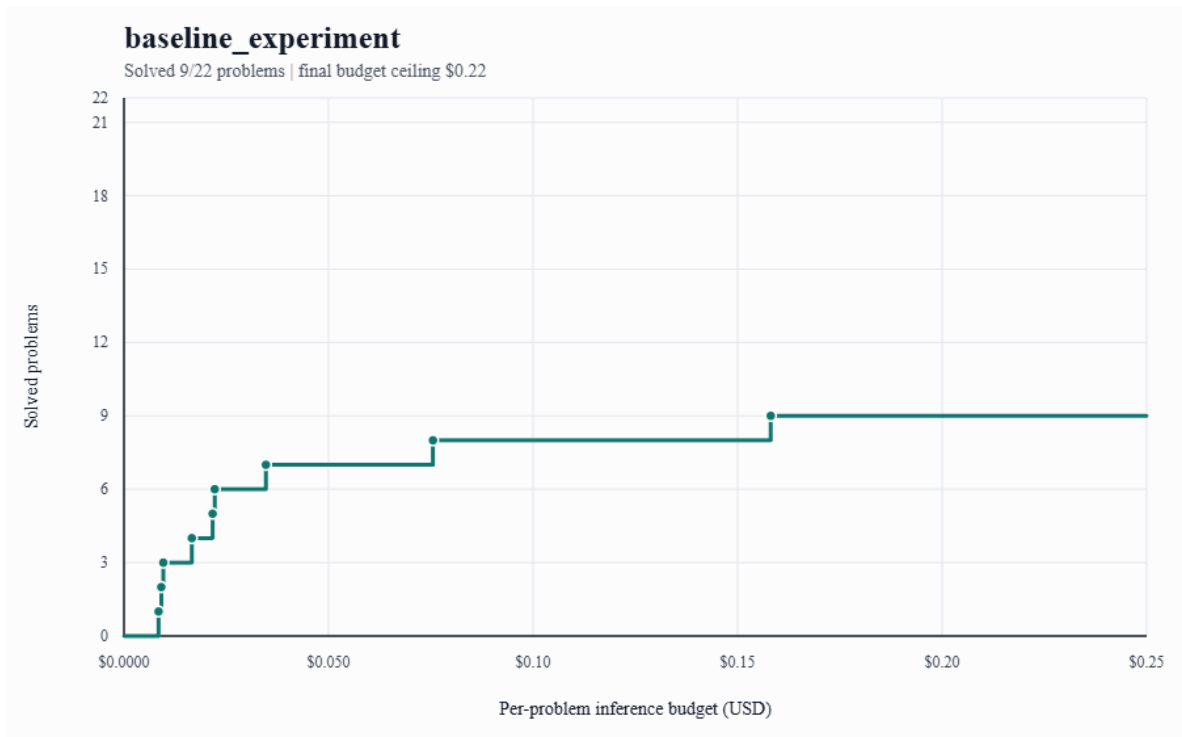
We observe similar patterns regarding to reasoning, although with an outlier.

“Medium” offers considerably better performance, ”low” is the outlier.

“High” moves again the curve to the left.

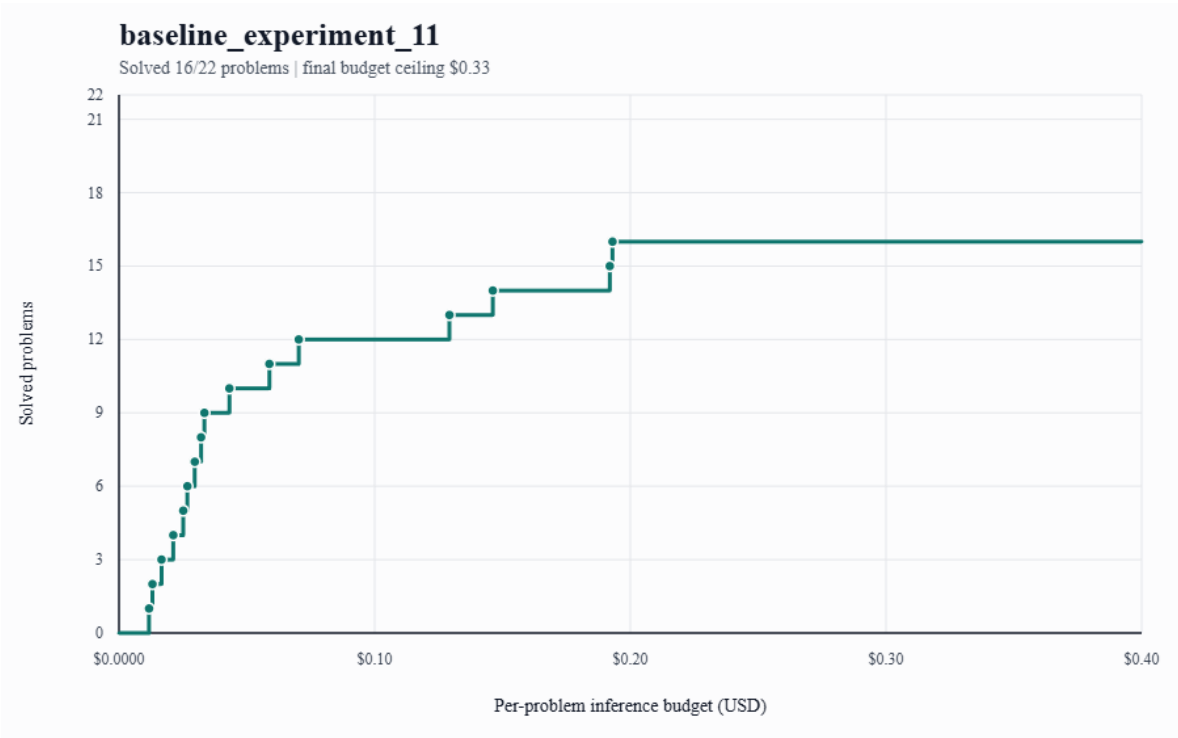
Baseline Experiment 10

Model	5.4-nano	Correct	9
Max_budget	0.2	Total cost	3,00
Max_loops	10		
Resoning effort	low		



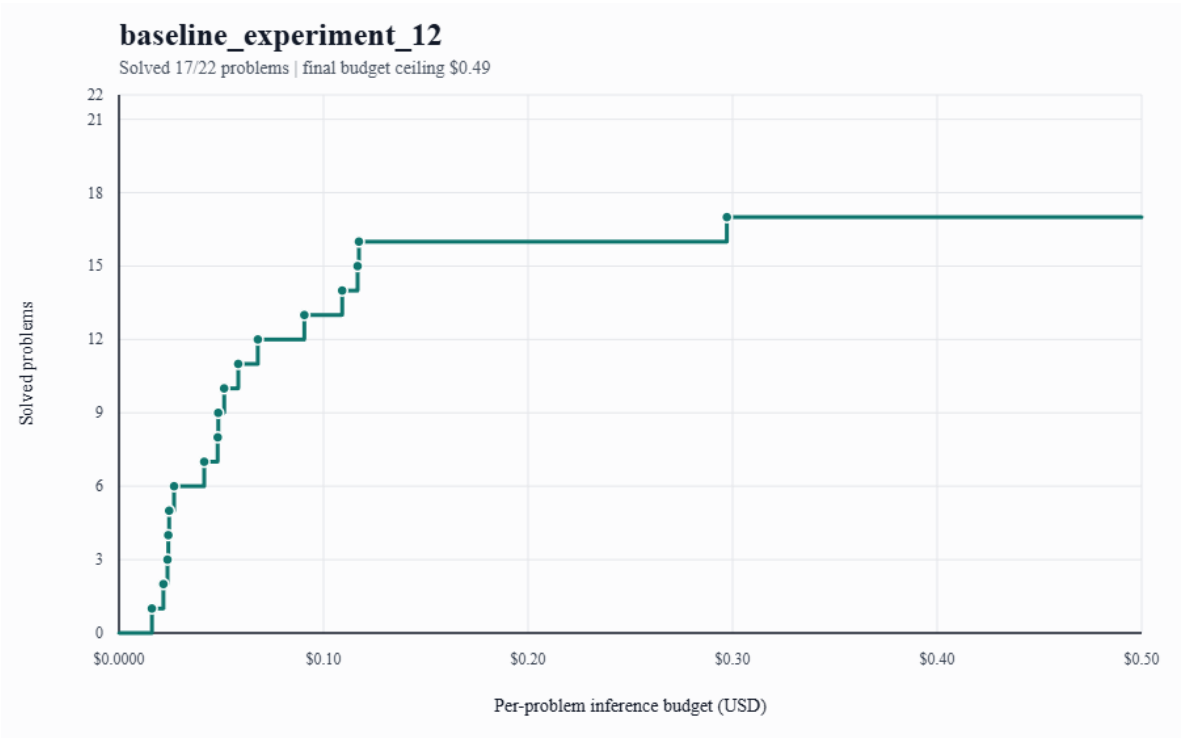
Baseline Experiment 11

Model	5.4-nano	Correct	16
Max_budget	0.2	Total cost	2,7
Max_loops	10		
Resoning effort	medium		



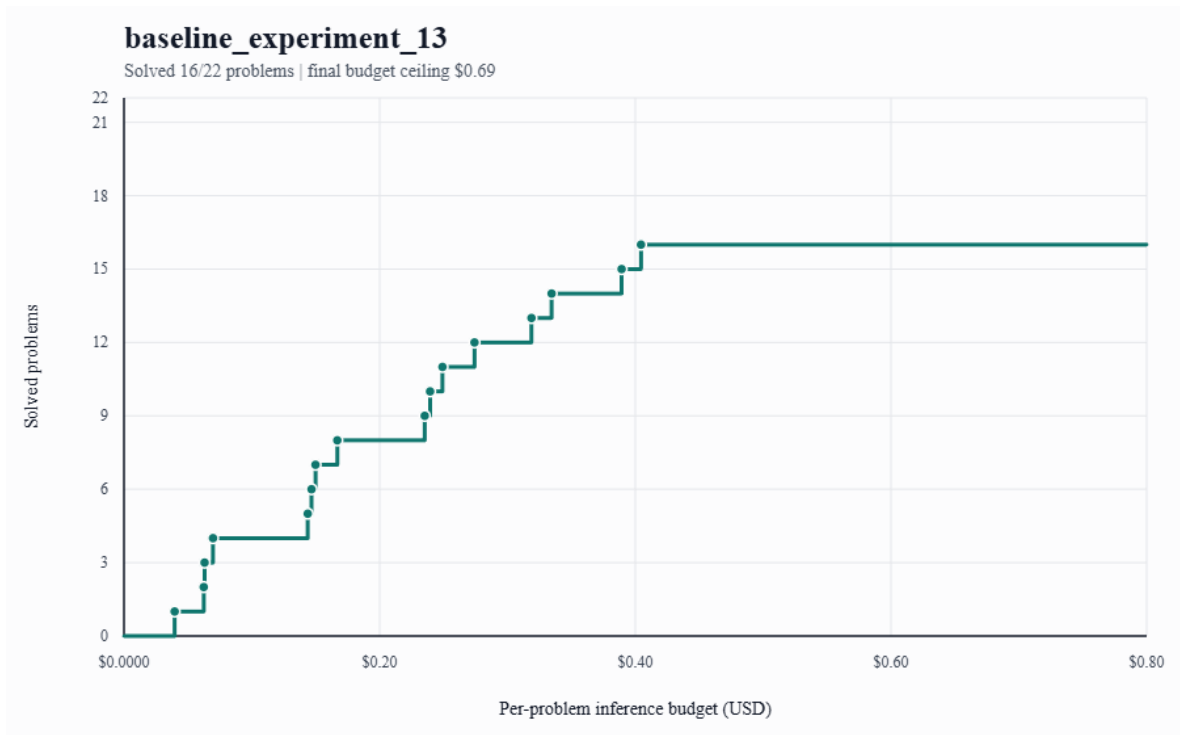
Baseline Experiment 12

Model	5.4-nano	Correct	17
Max_budget	0.2	Total cost	2,6
Max_loops	10		
Resoning effort	high		



Baseline Experiment 13

Model	5.4-nano	Correct	16
Max_budget	0.2	Total cost	6.5
Max_loops	10		
Resoning effort	xhigh		



Insight - mini vs nano

15 at 1,6\$ vs 11 at 2,9 \$ vs 9 at 3

19 at 1,13\$ vs 10 at 1,8 \$ vs 16 at 2,7

19 at 1,4\$ vs 14 at 3,3 \$ vs 17 at 2,6

Stronger contender than mini, but by doubling the price. Outliner -> Nano models can pack a punch -> **WHY?**

Insight – Reasoning effort

Here, it peaks at high, rather than [low, medium]

Eventhough this happens, [xhigh] is not worth pursuing -> 4 dollars more expensive, 1 problem less solved

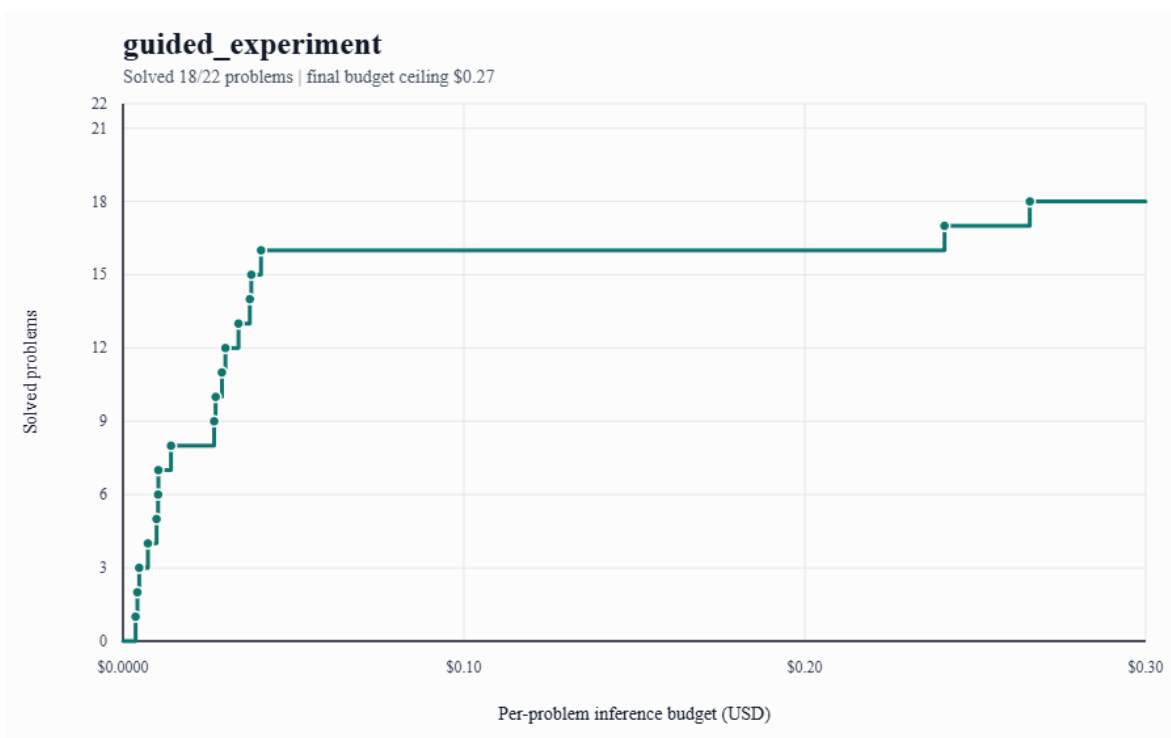
Nano + high reasoning effort -> surprinsinmgly good! Although expensive

Guided Experiment

As similar to the original baseline, we are going to tackle directly all models with higher reasoning efforts.

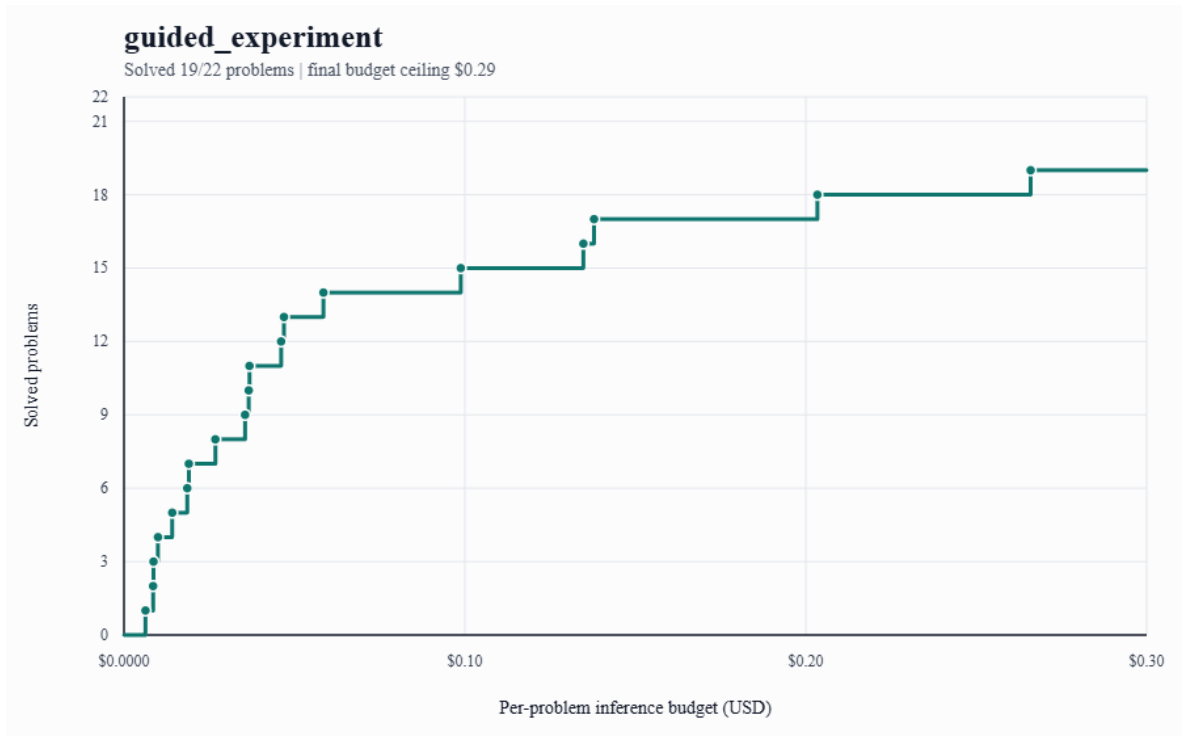
Guided experiment 14

Model	5.4	Correct	18
Max_budget	0.2	Total cost	1,7
Max_loops	10	N_comp	2
Resoning effort	low		



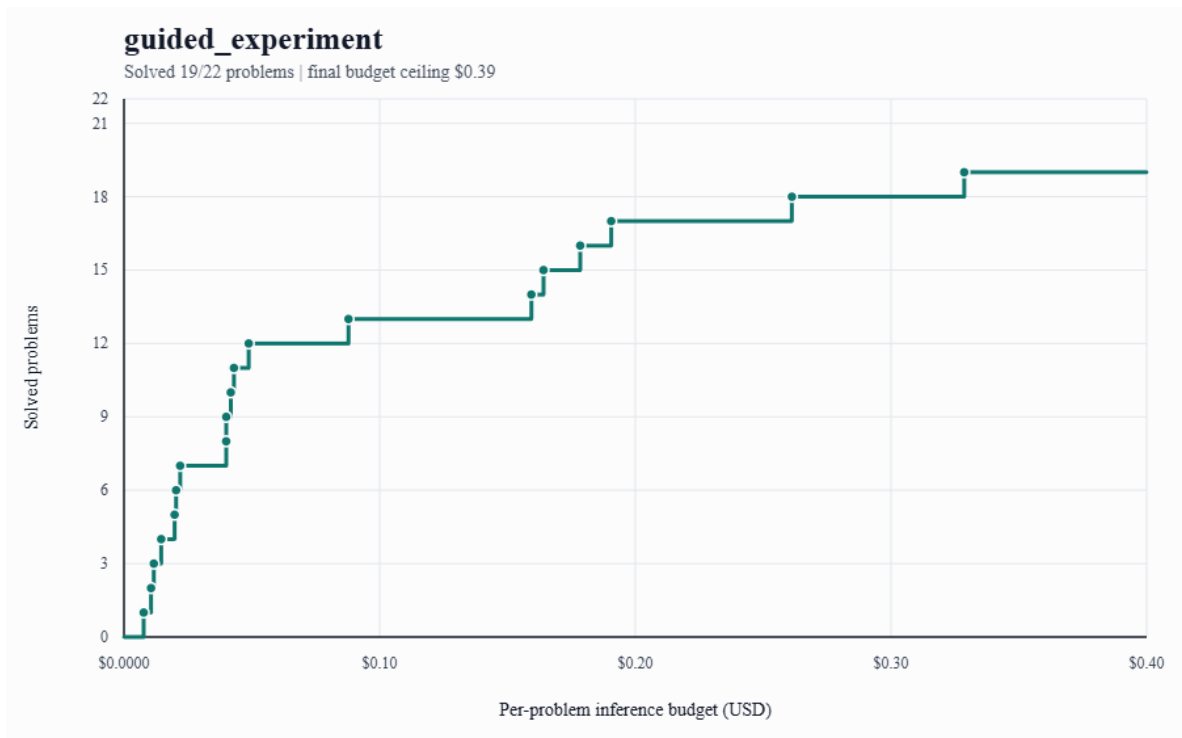
Guided experiment 15

Model	5.4	Correct	19
Max_budget	0.2	Total cost	2.00
Max_loops	10	N_comp	2
Resoning effort	medium		



Guided experiment 16

Model	5.4	Correct	19
Max_budget	0.2	Total cost	2,6
Max_loops	10	N_comp	2
Resoning effort	high		



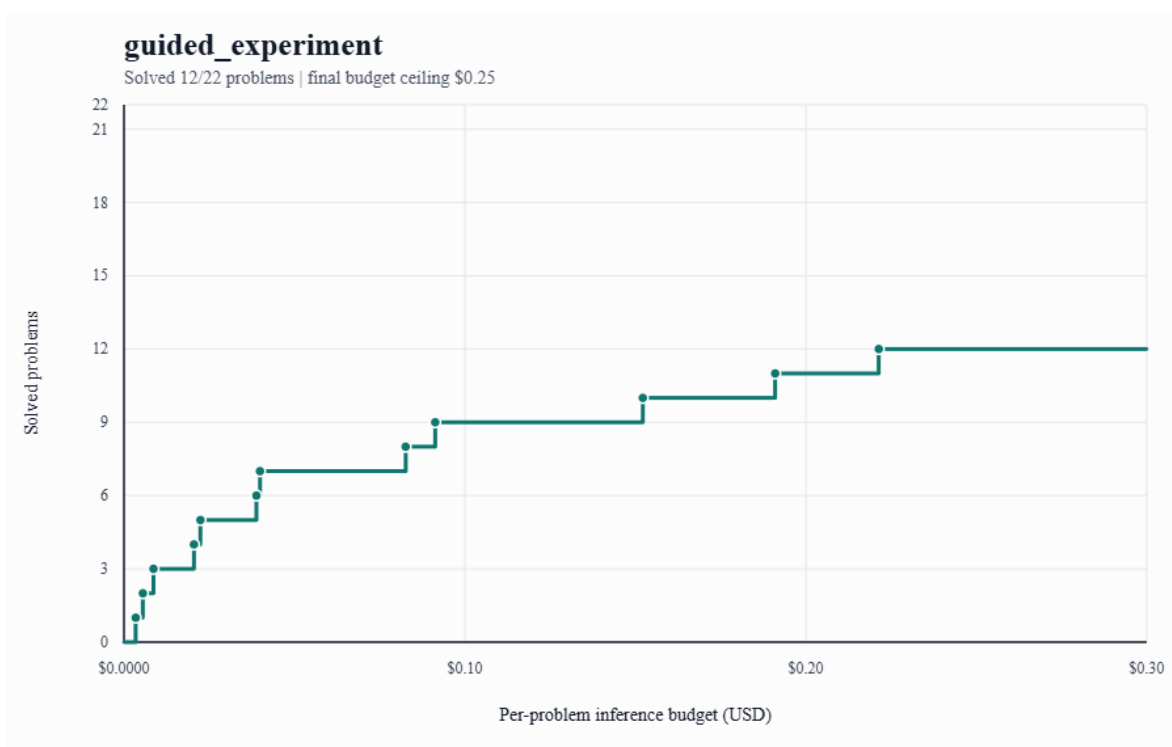
Insights

Really similar performance to baseline, although around 0.5 more expensive each time. High reasoning effort -> increases cost but does not decrease performance.

Guided experiment 17

Model	5.4-mini	Correct	12
Max_budget	0.2	Total cost	3,09
Max_loops	10	N_comp	2
Resoning effort	low		

Vs baseline, mini, low -> 10 at 2,87

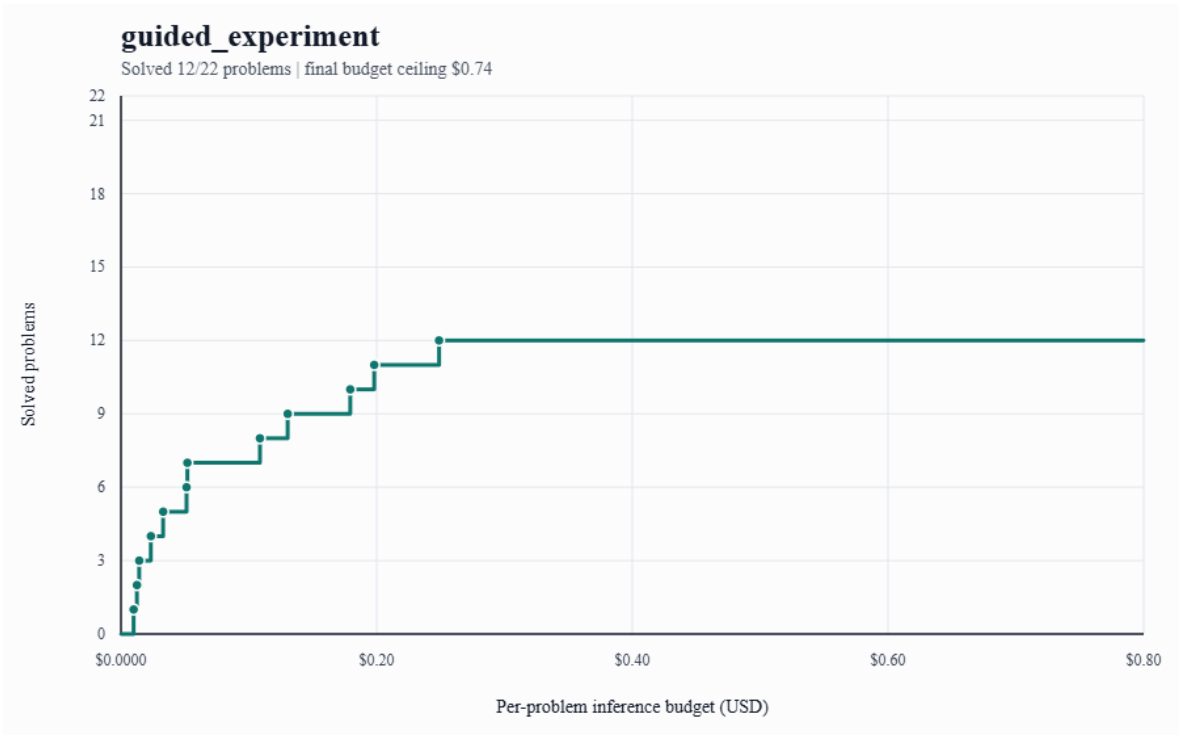


Guided experiment 18

Model	5.4-mini	Correct	12
Max_budget	0.2	Total cost	4,5
Max_loops	10	N_comp	2

Resoning effort	medium		
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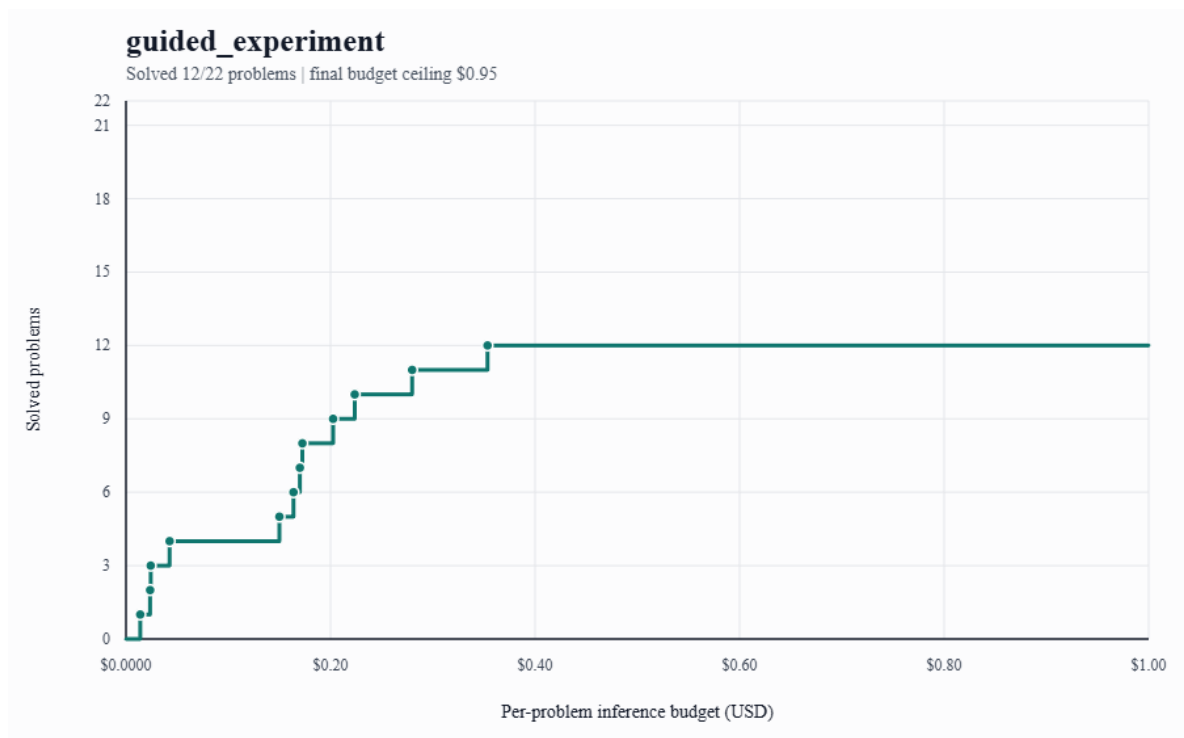
Vs baseline, mini, medium -> 14 at 3,3



Guided experiment 19

Model	5.4-mini	Correct	12
Max_budget	0.2	Total cost	8,3
Max_loops	10	N_comp	2
Resoning effort	high		

Vs baseline, mini, high -> 10 at 3,95

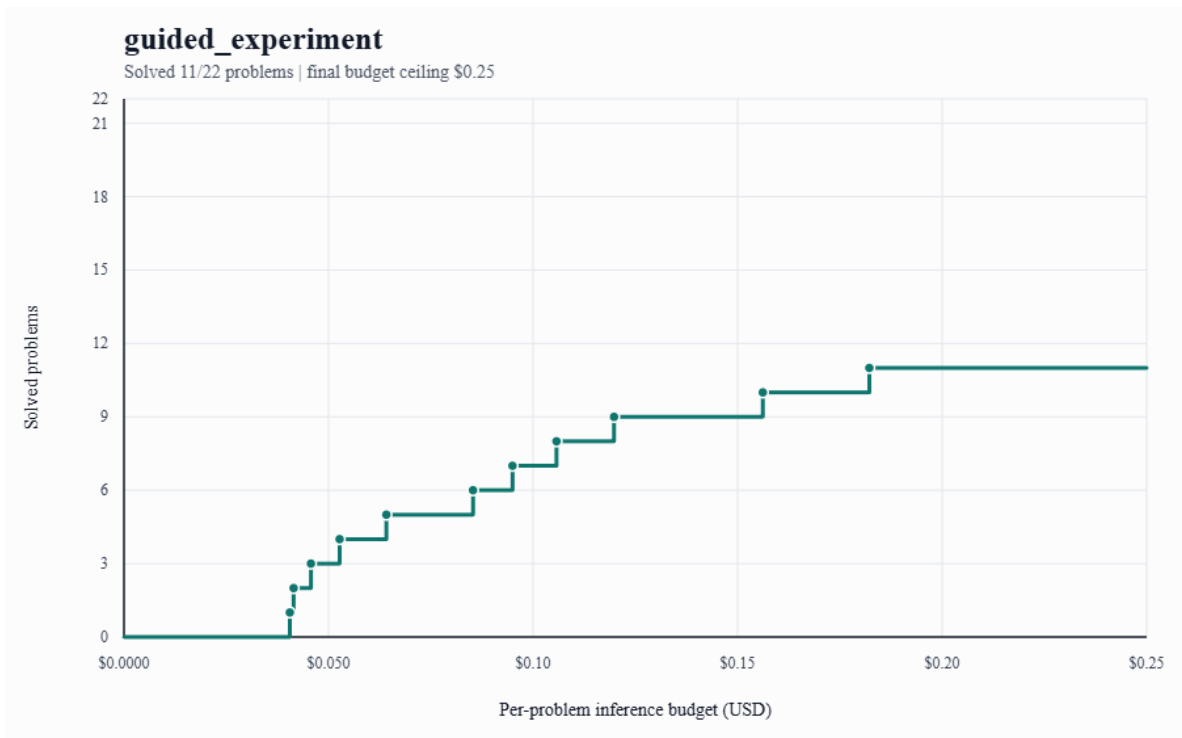


Insights here

Guided experiment 20

Model	5.4-nano	Correct	11
Max_budget	0.2	Total cost	3,4
Max_loops	10	N_comp	2
Resoning effort	low		

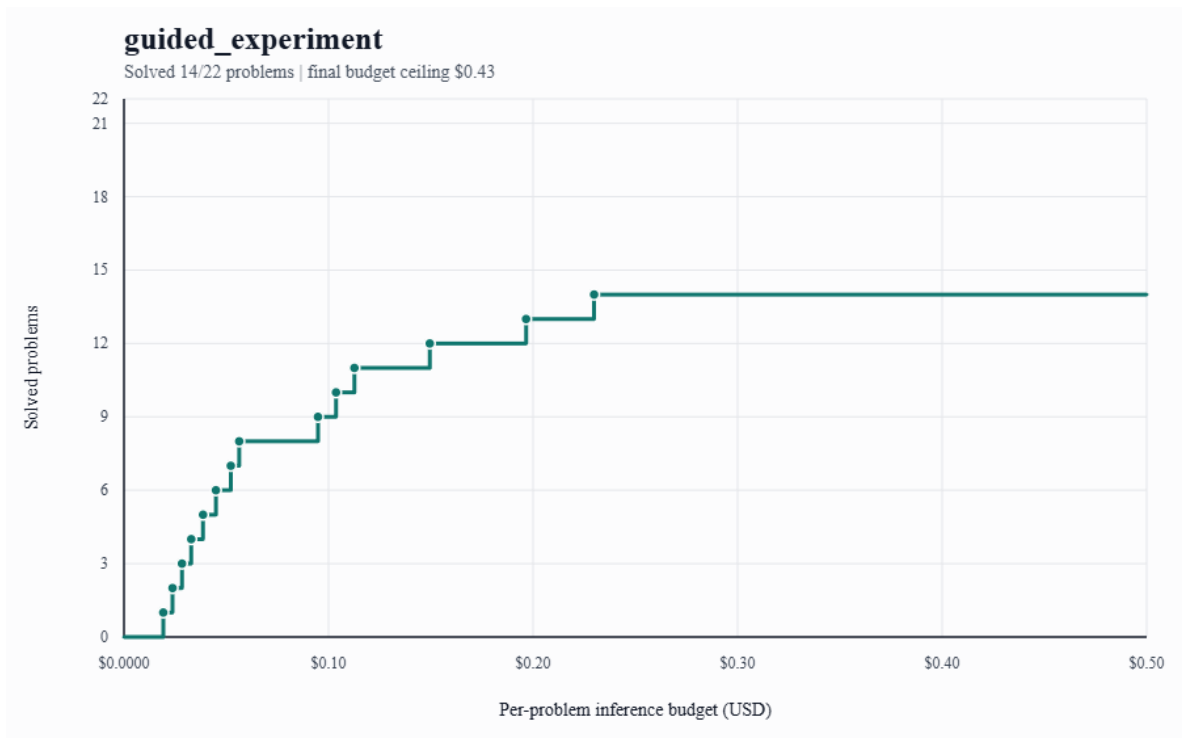
Vs baseline, nano, low -> 9 at 3,00



Guided experiment 21

Model	5.4-nano	Correct	14
Max_budget	0.2	Total cost	3,4
Max_loops	10	N_comp	2
Resoning effort	medium		

Vs baseline, nano, low -> 16 at 2,70



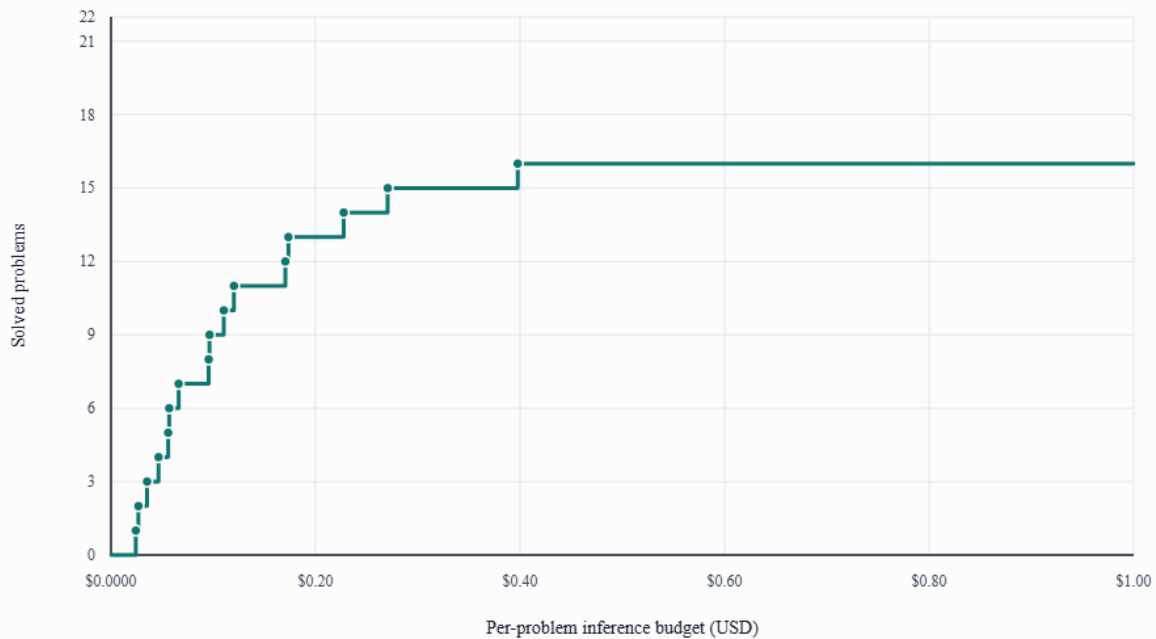
Guided experiment 22

Model	5.4-nano	Correct	16
Max_budget	0.2	Total cost	5
Max_loops	10	N_comp	2
Resoning effort	high		

Vs baseline, nano, hugh -> 17 at 2,6

guided_experiment

Solved 16/22 problems | final budget ceiling \$0.83

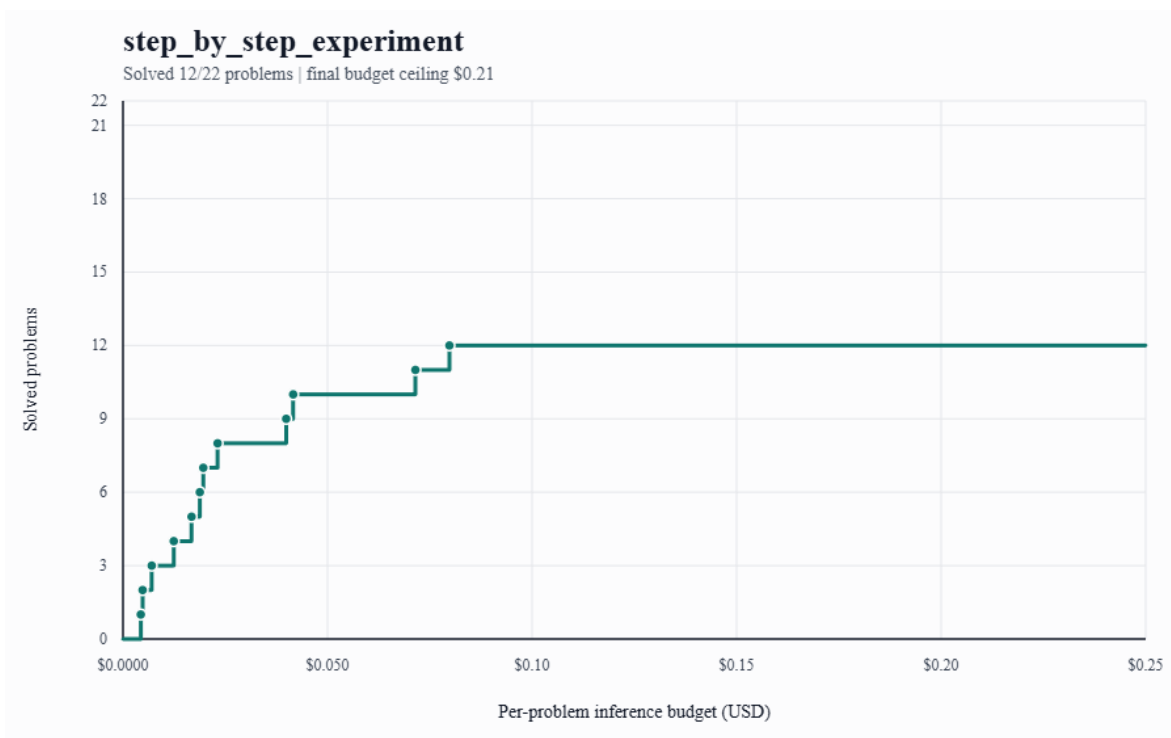


Step by Step Experiment

SBS 23

Sketch Model	5.4	Correct	12
Sketch effort	Low	Total cost	2,4
Step Model	Mini	N_comp	2
Step effort	low	Max_steps	12
Max_fb_per_step	3		
Max_budget	0.2		
Max_loops	10		

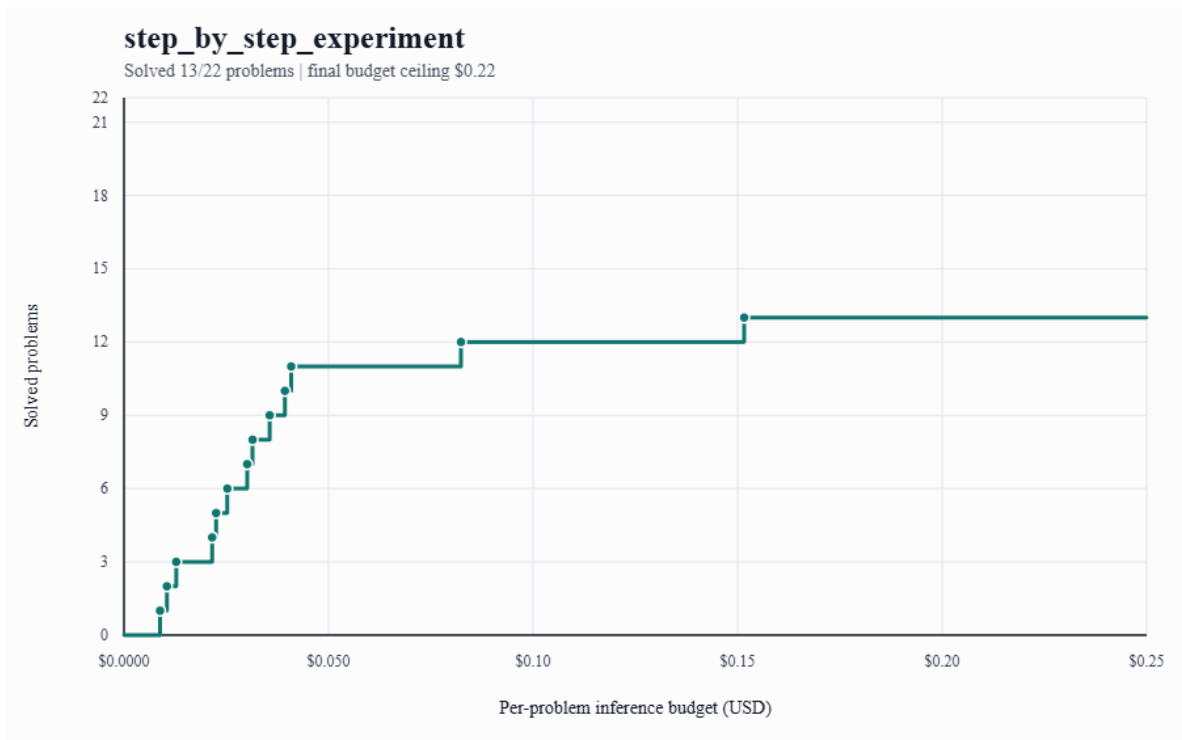
Vs baseline at 5.4 at low: 19 at 1.13 USD



SBS 24

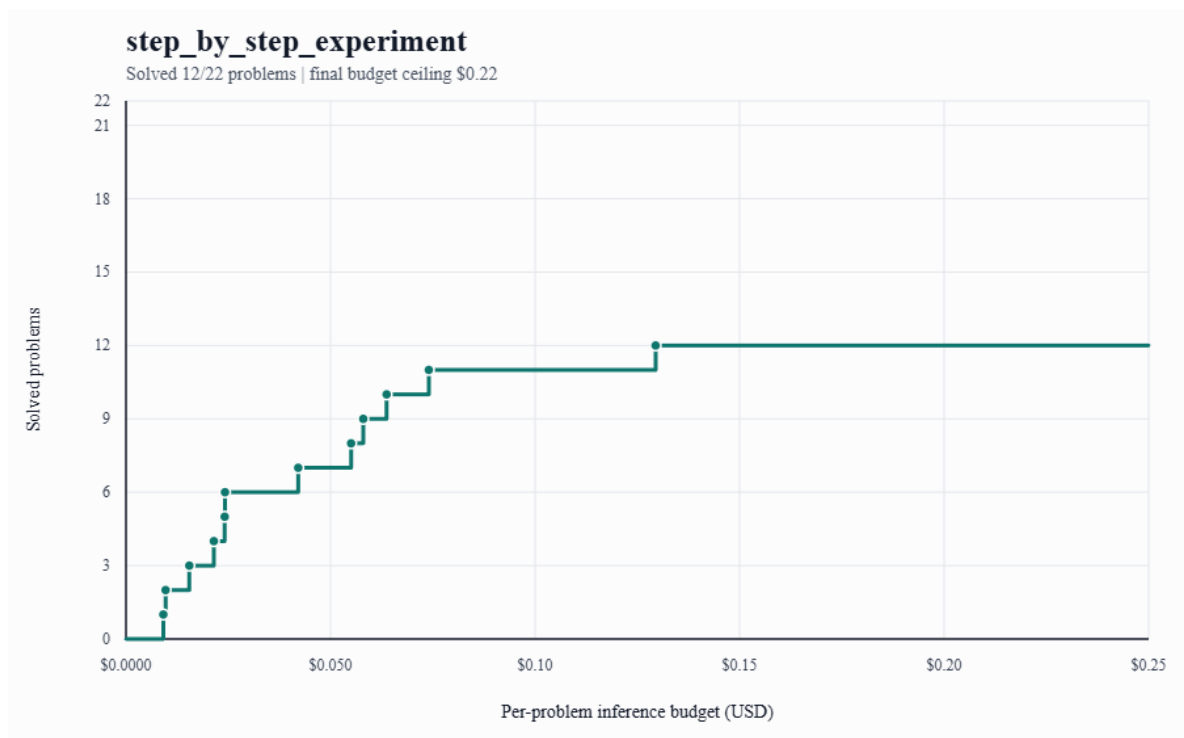
Sketch Model	5.4	Correct	13
Sketch effort	medium	Total cost	2,38
Step Model	Mini	N_comp	2
Step effort	low	Max_steps	12
Max_fb_per_step	3		
Max_budget	0.2		
Max_loops	10		

Vs baseline at 5.4 at low: 19 at 1.13 USD



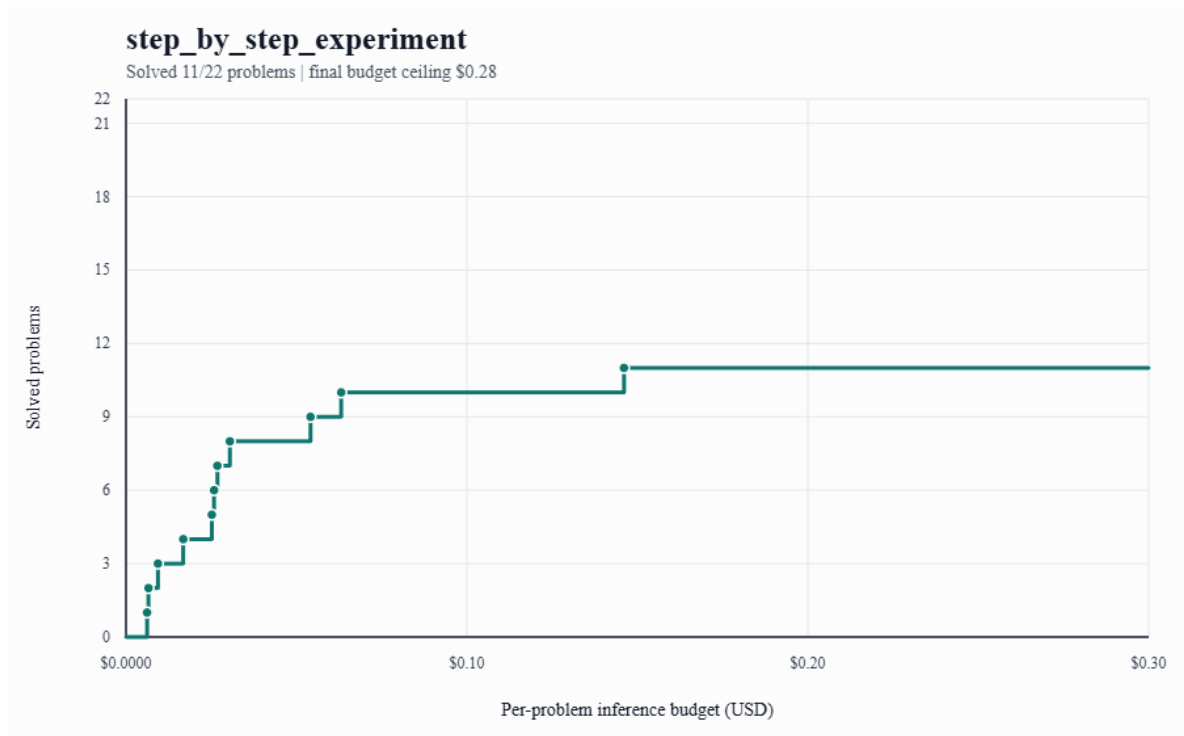
SBS 25

Sketch Model	5.4	Correct	12
Sketch effort	high	Total cost	2,6
Step Model	Mini	N_comp	2
Step effort	Low	Max_steps	12
Max_fb_per_step	3		
Max_budget	0.2		
Max_loops	10		



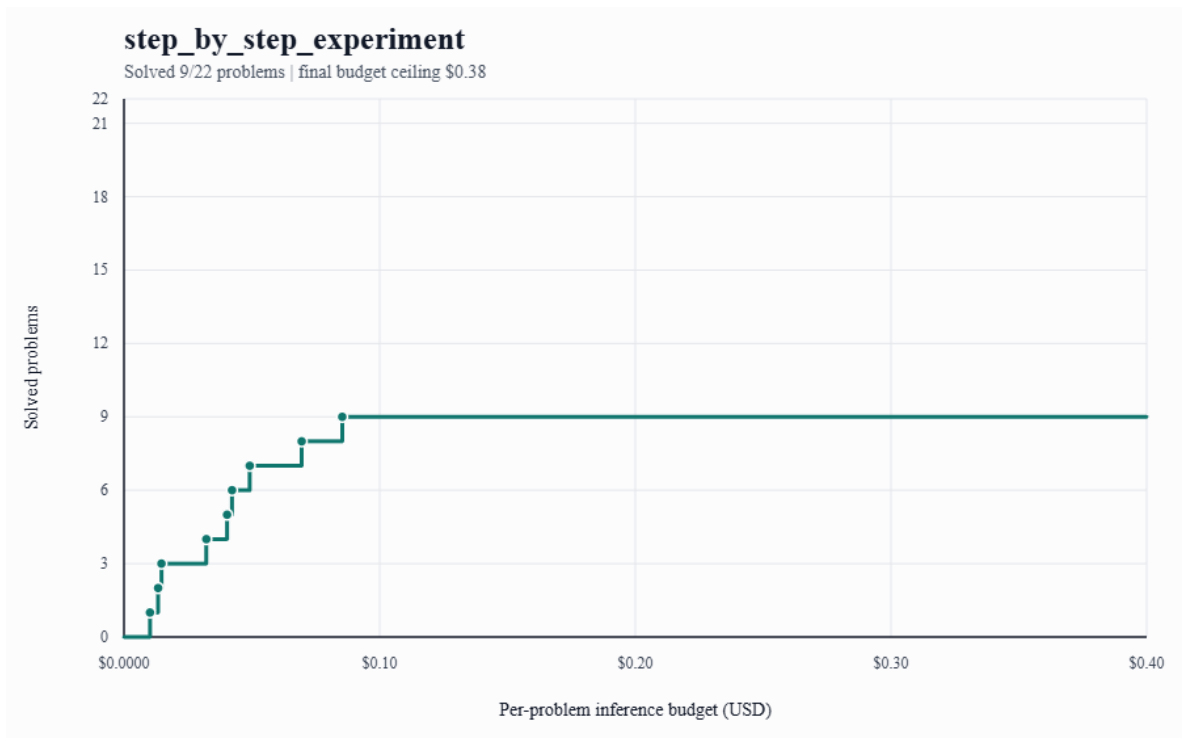
SBS 26

Sketch Model	5.4	Correct	11
Sketch effort	low	Total cost	2,8
Step Model	Mini	N_comp	2
Step effort	Medium	Max_steps	12
Max_fb_per_step	3		
Max_budget	0.2		
Max_loops	10		



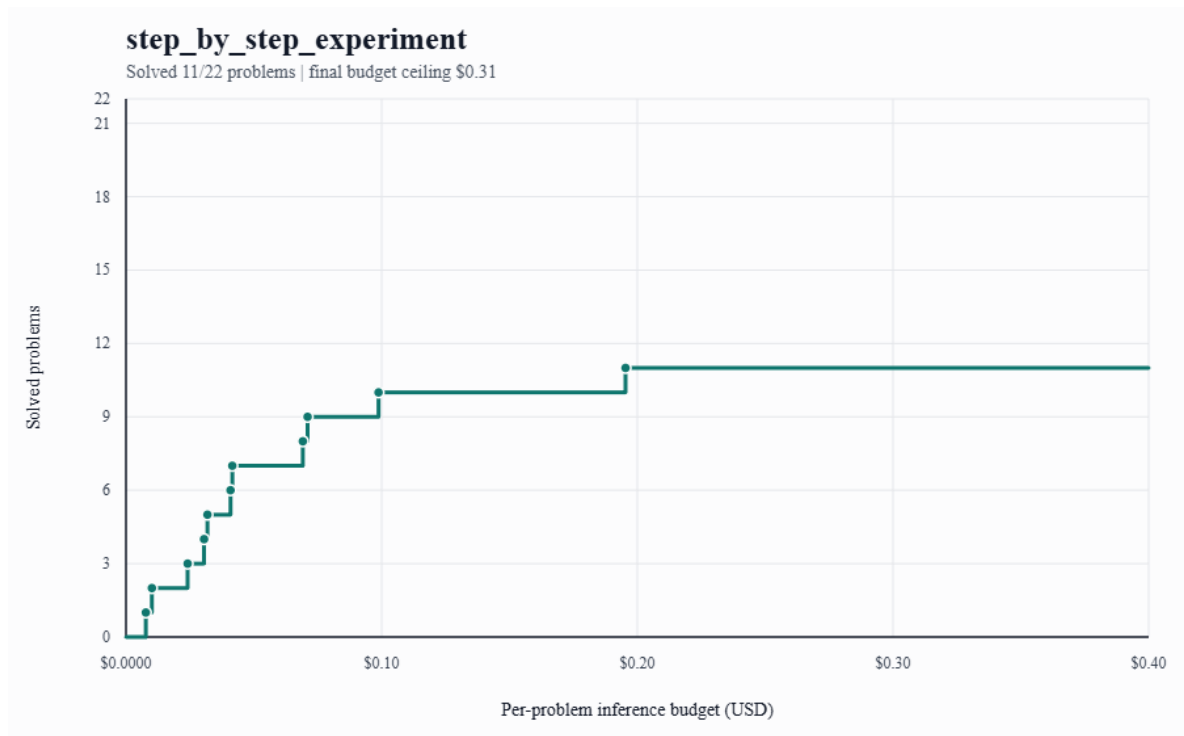
SBS 27

Sketch Model	5.4	Correct	9
Sketch effort	low	Total cost	4
Step Model	Mini	N_comp	2
Step effort	high	Max_steps	12
Max_fb_per_step	3		
Max_budget	0.2		
Max_loops	10		



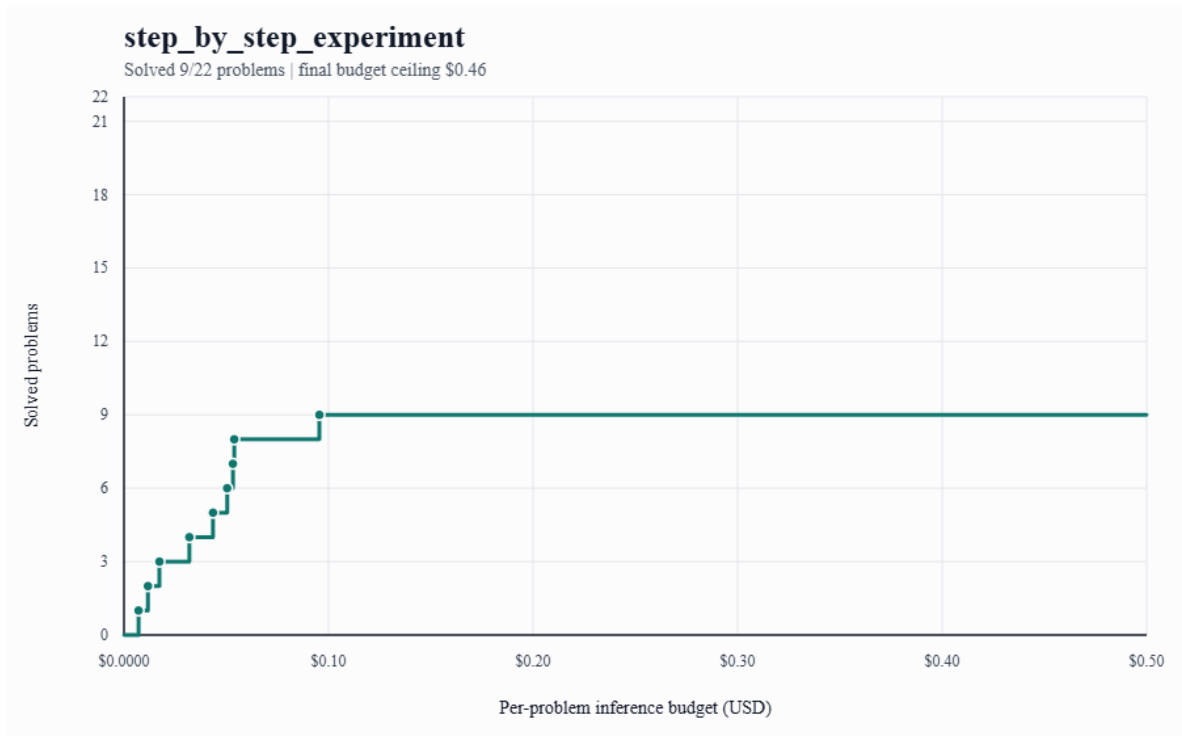
SBS 28

Sketch Model	5.4	Correct	11
Sketch effort	Medium	Total cost	3,2
Step Model	Mini	N_comp	2
Step effort	medium	Max_steps	12
Max_fb_per_step	3		
Max_budget	0.2		
Max_loops	10		



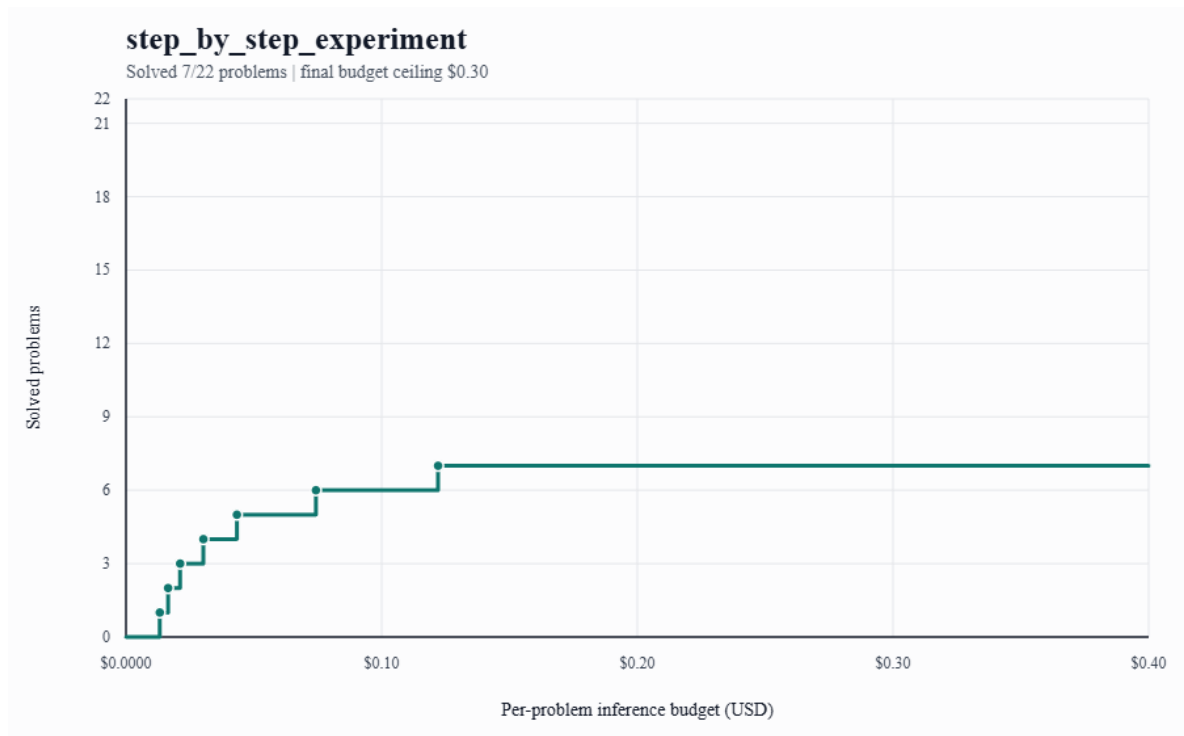
SBS 29

Sketch Model	5.4	Correct	9
Sketch effort	Medium	Total cost	3,8
Step Model	Mini	N_comp	2
Step effort	high	Max_steps	12
Max_fb_per_step	3		
Max_budget	0.2		
Max_loops	10		



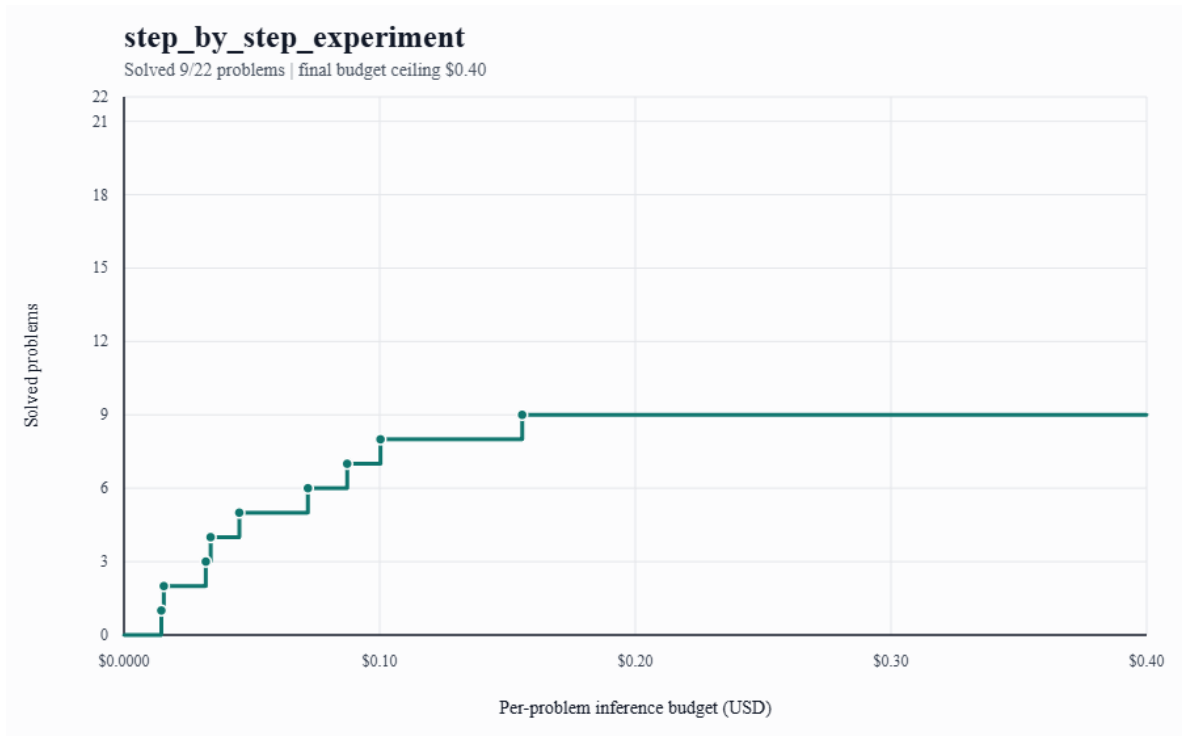
SBS 30

Sketch Model	5.4	Correct	7
Sketch effort	high	Total cost	3,9
Step Model	Mini	N_comp	2
Step effort	medium	Max_steps	12
Max_fb_per_step	3		
Max_budget	0.2		
Max_loops	10		



SBS 31

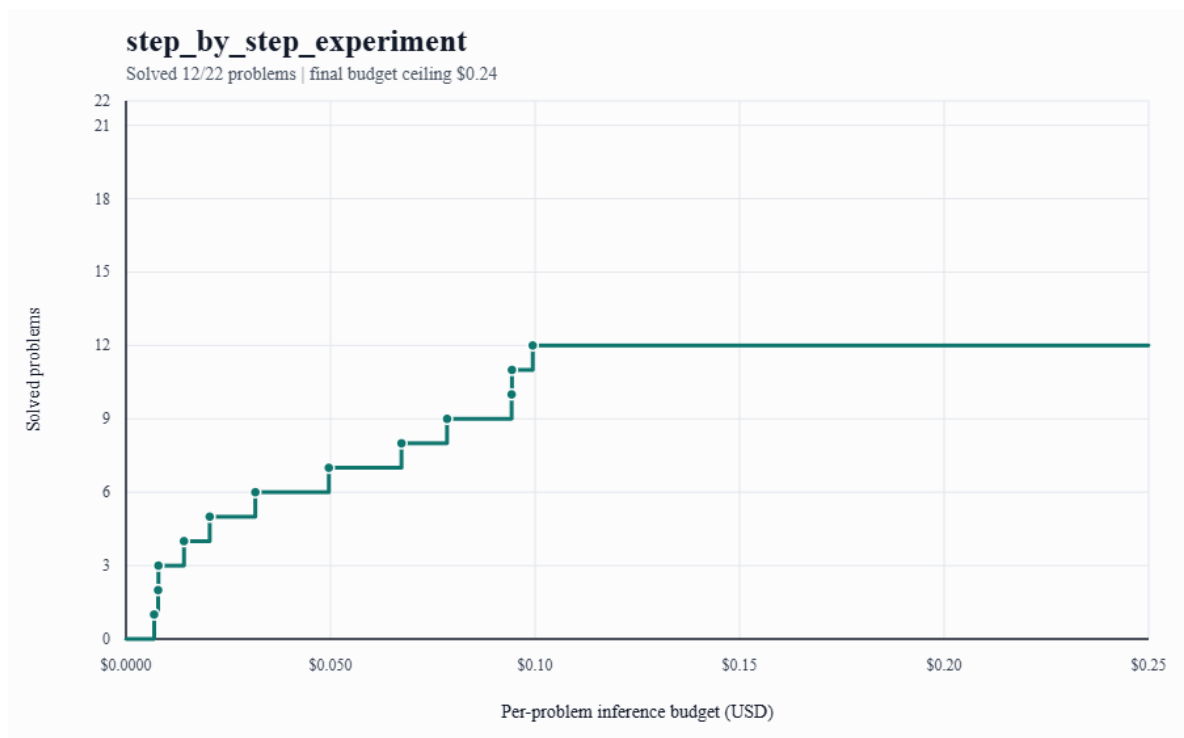
Sketch Model	5.4	Correct	9
Sketch effort	high	Total cost	3,8
Step Model	Mini	N_comp	2
Step effort	high	Max_steps	12
Max_fb_per_step	3		
Max_budget	0.2		
Max_loops	10		



As sketch best at medium, keep it that way and test nano

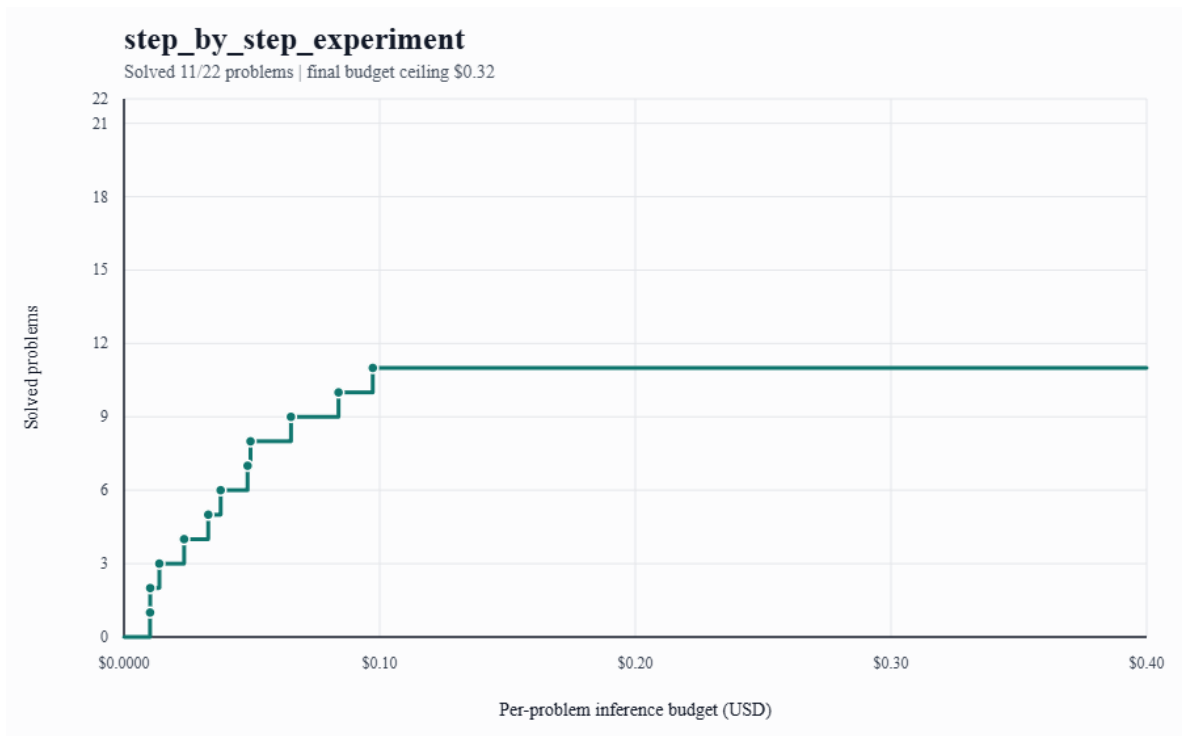
SBS 32

Sketch Model	5.4	Correct	12
Sketch effort	Medium	Total cost	2,7
Step Model	Nano	N_comp	2
Step effort	low	Max_steps	12
Max_fb_per_step	3		
Max_budget	0.2		
Max_loops	10		



SBS 33

Sketch Model	5.4	Correct	11
Sketch effort	Medium	Total cost	3
Step Model	Nano	N_comp	2
Step effort	medium	Max_steps	12
Max_fb_per_step	3		
Max_budget	0.2		
Max_loops	10		



SBS 34

Sketch Model	5.4	Correct	
Sketch effort	Medium	Total cost	
Step Model	Nano	N_comp	2
Step effort	high	Max_steps	12
Max_fb_per_step	3		
Max_budget	0.2		
Max_loops	10		